DME, Done Right

Domestic dimethyl ether for LPG blending — commercial segment first, energy-parity pricing always

Investment pitch · July 2026 · built on the Volume Dividend fourth-quadrant analysis, LERC engineering evidence and BIS IS 18698:2024

The setup: a huge import bill and a freshly opened standard

...and a trap that will kill naive DME programmes

₹90,000+ cr of LPG imports

India imports ~62% of its 29.7-MMT LPG pool — propane/butane priced off the Saudi CP, dollar-denominated, geopolitically exposed. Every blending programme starts from this bill.

The door is open

BIS IS 18698:2024 permits up to 20% DME in LPG. LERC (LPG Equipment Research Centre, Bengaluru) has cleared 20% material compatibility and run stable flex-burner trials — the engineering is done.

The trap (our own analysis)

DME carries 37% less energy per kg. On subsidised domestic cylinders at unchanged prices, DME20 bills households +₹312/yr AND adds ₹1,201 cr/yr of PMUY subsidy — a programme designed that way dies politically. The pitch is built NOT to be that programme.

- LERC's measured thermal-efficiency drop at DME20 is −5.26% (IS 4246) — real but manageable IF the pricing is honest; the venture's entire design flows from that condition

The guardrails ARE the strategy

Three self-imposed conditions that make the venture durable where naive DME dies

1 • Energy-parity pricing, always

DME sells at LPG price × 28.8/45.8 (~0.63× per kg). The buyer pays for energy, not kilograms. This forgoes the ₹34.7/kg 'temptation margin' — which is precisely the consumer's hidden loss and the programme's political poison.

2 • Commercial LPG first

The 19-kg commercial cylinder is price-DEREGULATED and unsubsidised: parity pricing is a private negotiation, not a subsidy fight. The PMUY inversion never arises. Domestic kitchens only after parity cylinder pricing is regulated in.

3 • Domestic carbon only

Imported-methanol DME earns ₹1.9/kg at honest pricing — it fails, and merely re-denominates the import bill. The venture runs on domestic coal-gasification / bio-methanol under long-term contract, or it does not run.

- Honesty as moat: a venture that cannot be accused of the fourth-quadrant harm is the one regulators let scale — the same durability logic as CBG's on-budget stack

The wedge: commercial & industrial LPG

Unsubsidised, deregulated, price-sensitive buyers — hotels, restaurants, industry

Segment	Pool (MMT/yr)	DME20 ceiling (kt DME)	Why it works
Commercial (19-kg) + industrial	~4.5	~900	deregulated; parity = commercial discount
Auto-LPG	~0.3	~60	niche; elastomer retrofit known
Domestic (14.2-kg, PMUY)	~25	—	EXCLUDED until parity pricing is regulated

The commercial buyer's math

- DME20 at energy parity cuts the effective ₹/MJ of a hotel's fuel bill ~2–3% (blender passes part of the DME discount)
- Same burners; elastomer swap per LERC compatibility list

Ceiling with today's standard

- C&I DME20 ceiling ≈ 0.9–1.1 MMT DME/yr = 27–33 plants of the reference size — years of runway before the domestic segment is even needed

Business model: the simple step in the chain

100-TPD methanol-to-DME dehydration plant, co-located with domestic methanol capacity

Domestic methanol

coal-gasification / bio-methanol, long-term contract ₹15–17/kg

→

Dehydration unit

catalytic (alumina) — proven, simple; the ₹120 cr asset

→

DME storage & loading

LPG-like handling; pressurised spheres

→

Blender / marketer

OMC & parallel marketers blend ≤20% per IS 18698

→

C&I customer

19-kg cylinders & bulk — energy-parity billed

Per-kg economics at the three postures (₹/kg)

	Realisation	Methanol feed (1.4×)	Conversion	Margin
Base: domestic methanol, parity	39.0	(23.8)	(3.5)	11.7
Imported methanol (fails)	39.0	(33.6)	(3.5)	1.9
kg-parity pricing (refused)	62.0	(23.8)	(3.5)	34.7

Financials: honest pricing still clears 26%

₹120 cr capex · 70:30 @ 10% (no subvention assumed) · 29.7 kt DME/yr · energy-parity realisation ₹39/kg

26.0%	42.1%	₹106 cr	Year 4	1.94×
Project IRR	Equity IRR	NPV @ 12%	Payback	Steady DSCR

Sensitivity — project IRR (%): methanol price × DME realisation

Methanol ₹/kg	₹36/kg	₹39/kg (parity, base)	₹42/kg
15,000/t	25.6	32.4	39.0
17,000/t (base)	18.8	26.0	32.9
20,000/t	6.7	15.6	23.2

The whole model rides on the methanol contract: ₹20/kg methanol compresses IRR to 15.6% — the feed contract IS the investment decision

- No subvention assumed — a DME-specific interest-subvention analogue (the ethanol ISS template) is pure upside; EBITDA ₹34.7 cr/yr at base

Risks and mitigants

Named in our own fourth-quadrant analysis — answered structurally

Risk	Reality	Mitigant
Methanol supply	Domestic coal/bio-methanol capacity is nascent; market methanol is imported ₹24	Co-location + take-or-pay with Coal India/BHEL-route projects; venture gated on the contract, not built ahead of it
Elastomer retrofit	DME attacks conventional seals beyond ~20%	Stay ≤20% per IS 18698; LERC compatibility list is the binding spec; retrofit kits funded at blender
Demand adoption	C&I buyers must accept a blended product	Energy-parity billing makes the buyer strictly better off; pilot with 2 parallel marketers
Policy drift	Pressure to push into subsidised domestic LPG at kg prices	Charter commitment: domestic segment only under regulated parity pricing — the refusal is contractual
Saudi CP swings	Parity realisation moves with LPG	Margin is methanol-linked, not CP-linked, at ~63% pass-through; hedge on the spread

- Not relying on: subvention, carbon credits, domestic-kitchen volumes, or any pricing above energy parity

Landscape: technology proven, field empty

China blends DME-LPG at scale; India has the standard but no supply

Global precedent

China has blended DME into LPG at multi-MMT scale for two decades (the cautionary tales are all kg-parity mis-selling — our guardrail #1 exists because of them). Korea/Japan ran fleet and burner programmes.

Indian building blocks

NCL Pune has demonstrated DME synthesis; IOCL R&D active; Assam Petrochemicals and GNFC hold methanol capacity; Coal India–BHEL coal-to-methanol projects are the feed pipeline. Nobody has assembled the chain.

First-mover reality

India has a 20% standard (IS 18698:2024), LERC-cleared equipment evidence, a ₹90,000-cr import bill — and zero commercial DME blending. The venture IS the market at inception.

- Synergy with the bioenergy platform: bio-methanol/bio-DME from CBG-adjacent syngas is the long-run domestic-carbon route — same feedstock ecosystems, same partners

The ask: one gated pilot, then replicate

₹120 cr	29.7 kt/yr	26% / 42%	27–33×
pilot plant (₹36 cr equity)	DME = ~150 kt DME20 comm. LPG	project / equity IRR at parity	replication ceiling in C&I alone

- Gate 1 (pre-capex): 10-yr take-or-pay domestic methanol contract at ≤₹17/kg — no contract, no build; site co-located with the methanol plant
- Gate 2: blending MoU with two parallel marketers / one OMC for C&I DME20 at energy-parity transfer pricing; LERC-listed retrofit kits at the bottling plant
- Build (15 months) → prove the honest-pricing P&L publicly → replicate toward the ~30-plant C&I ceiling; domestic kitchens only when parity cylinder pricing is regulated
- Why fund the constrained version: the unconstrained margin (₹34.7/kg at kg-parity) is exactly what makes naive DME uninvestable — this structure is the one that survives scrutiny, scale and time

Model & sensitivity: dme_pitch_model.py · analysis: dme_lpg_blending.py (LERC basis) · The Volume Dividend §10 · github.com/herrrickshaw/india-omc-fuel-fleet-model